STACKS, MODULI OF CURVES: LEMMA-EXTEND-CURVES-TO-SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 298-305 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 307-362, the complete author proof (54 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_moduli-curves.json (pb=2671, stmt 298-305, proof 307-362).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0ADV: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D4Z: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041V: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 041W: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0422: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 044U: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06LR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 362–362. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $T \to B$ be a morphism of algebraic spaces. The category*

$$\mathrm{Mor}_B(T, B\text{-}\mathit{Curves}) = \mathrm{Mor}(T, \mathit{Curves})$$

is the category of families of curves over T.

Proof. A family of curves over T is a morphism $f : X \to T$ of algebraic spaces, which is flat, proper, of finite presentation, and has relative dimension ≤ 1 (Morphisms of Spaces, Definition 06LR). This is exactly the same as the definition in Quot, Situation 0D4Z except that T the base is allowed to be an algebraic space. Our default base category for algebraic stacks/spaces is the category of schemes, hence the lemma does not follow immediately from the definitions. Having said this, we encourage the reader to skip the proof.

By the product description of B-*Curves* given above, it suffices to prove the lemma in the absolute case. Choose a scheme U and a surjective étale morphism $p : U \to T$. Let $R = U \times_T U$ with projections $s, t : R \to U$.

Let $v : T \to \mathit{Curves}$ be a morphism. Then $v \circ p$ corresponds to a family of curves $X_U \to U$. The canonical 2-morphism $v \circ p \circ t \to v \circ p \circ s$ is an isomorphism $\varphi : X_U \times_{U,s} R \to X_U \times_{U,t} R$. This isomorphism satisfies the cocycle condition on $R \times_{s,t} R$. By Bootstrap, Lemma 0ADV we obtain a morphism of algebraic spaces $X \to T$ whose pullback to U is equal to X_U compatible with φ. Since $\{U \to T\}$ is an étale covering, we see that $X \to T$ is flat, proper, of finite presentation by Descent on Spaces, Lemmas 041W, 0422, and 041V. Also $X \to T$ has relative dimension ≤ 1 because this is an étale local property. Hence $X \to T$ is a family of curves over T.

Conversely, let $X \to T$ be a family of curves. Then the base change X_U determines a morphism $w : U \to \mathit{Curves}$ and the canonical isomorphism $X_U \times_{U,s} R \to X_U \times_{U,t} R$ determines a 2-arrow $w \circ s \to w \circ t$ satisfying the cocycle condition. Thus a morphism $v : T = [U/R] \to \mathit{Curves}$ by the universal property of the quotient $[U/R]$, see Groupoids in Spaces, Lemma 044U. (Actually, it is much easier in this case to go back to before we introduced our abuse of language and direct construct the functor $\mathit{Sch}/T \to \mathit{Curves}$ which "is" the morphism $T \to \mathit{Curves}$.)

We omit the verification that the constructions given above extend to morphisms between objects and are mutually quasi-inverse. □